

Web Development

Labo 11

Theorie: CSS Grid

2.1 Grid container en grid items

Een HTML-element wordt een grid container door de CSS-eigenschap:

```
display: grid;
```

Alle directe kinderen van dit element worden automatisch grid items. Enkel deze directe kinderen kunnen gepositioneerd worden met grid-eigenschappen.

2.2 Definitie van rijen en kolommen

De structuur van een grid wordt bepaald met:

- `grid-template-columns`
- `grid-template-rows`

Voorbeeld:

```
grid-template-columns: 1fr 1fr 1fr;  
grid-template-rows: auto auto;
```

De eenheid fr (fractional unit) verdeelt de beschikbare ruimte proportioneel over de kolommen of rijen.

2.3 Automatisch gegenereerde rijen

Wanneer er meer grid items zijn dan expliciet gedefinieerde rijen, genereert de browser automatisch extra rijen. Deze rijen krijgen standaard de hoogte auto, tenzij anders ingesteld met `grid-auto-rows`.

2.4 Positionering van grid items

Grid items kunnen expliciet gepositioneerd worden met:

- `grid-column`
- `grid-row`

Voorbeeld:

```
grid-column: 2 / 4;  
grid-row: 1 / 3;
```

Dit betekent dat het element start in kolom 2 en eindigt vóór kolom 4.

Alternatief kan gebruik gemaakt worden van `span`:

```
grid-column: span 2;
```

2.5 Ruimte tussen grid items

Witruimte tussen grid items wordt voorzien met de eigenschap:

```
gap: 20px;
```

Dit is de correcte manier om ruimte tussen blokken te creëren binnen een grid en vervangt het gebruik van margins.

2.6 Responsive grids en media queries

CSS Grid werkt uitstekend samen met media queries. Hierdoor kan de lay-out volledig aangepast worden aan verschillende schermgroottes zonder HTML te wijzigen.

Voorbeeld:

```
@media (max-width: 700px) {  
  grid-template-columns: repeat(3, 1fr);  
}
```

2.7 Debugging

Via de Chrome Developer Tools kan de grid-structuur visueel weergegeven worden in de Layout-tab. Dit is een belangrijk hulpmiddel bij het analyseren en debuggen van complexe grids.

Opdracht 1

Aantal rijen

Er worden automatisch drie rijen gegenereerd.

Verklaring

Omdat er negen grid items zijn en slechts drie kolommen gedefinieerd worden, vult de browser elke rij met drie items. Zodra een rij vol is, wordt automatisch een nieuwe rij aangemaakt.

HTML	CSS
<pre> <!DOCTYPE html> <html lang="en"> <head> <title>Grid Opdracht 1</title> <meta charset="UTF-8"> <meta name="viewport" content="width=device-width, initial-scale=1"> <link rel="stylesheet" href="../CSS/default.css"> <link rel="stylesheet" href="../CSS/Opdracht%201%20CSS%20Grid.css"> </head> <body> <main> <header>Grid Opdracht 1</header> <nav> <div>&lt; Terug</div> </nav> <div class="container"> <div class="block">Block 1</div> <div class="block">Block 2</div> <div class="block">Block 3</div> <div class="block">Block 4</div> <div class="block">Block 5</div> <div class="block">Block 6</div> <div class="block">Block 7</div> <div class="block">Block 8</div> <div class="block">Block 9</div> </div> </main> </body> </html> </pre>	<pre> .container{ background- color: cornflowerblue; display: grid; grid-template- columns: auto auto auto; } .container .block{ margin:50px; padding:50px; background- color: white; } </pre>

Grid Opdracht 1

< Terug

Block 1

Block 2

Block 3

Block 4

Block 5

Block 6

Block 7

Block 8

Block 9

Opdracht 2

Weergave van de blokken

De blokken worden van links naar rechts en van boven naar beneden geplaatst. Omdat er twaalf grid-cellen beschikbaar zijn en slechts negen items, blijven drie cellen leeg.

HTML	CSS
<pre><!DOCTYPE html> <html lang="en"> <head> <title>Grid Opdracht 2</title> <meta charset="UTF-8"> <meta name="viewport" content="width=device-width, initial-scale=1"> <link rel="stylesheet" href="../CSS/default.css"> <link rel="stylesheet" href="../CSS/Opdracht%202%20CSS%20Grid.css"> </head> <body> <main> <header>Grid Opdracht 2</header> <nav> <div>&lt; Terug</div> </nav> <div class="container"> <div class="block">Block 1</div> <div class="block">Block 2</div> <div class="block">Block 3</div> <div class="block">Block 4</div> <div class="block">Block 5</div> <div class="block">Block 6</div> <div class="block">Block 7</div> <div class="block">Block 8</div> <div class="block">Block 9</div> </div> </main> </body> </html></pre>	<pre>.container{ background-color: cornflowerblue; display: grid; grid-template-columns: auto auto auto auto; grid-template-rows: auto auto auto; } .container .block{ margin:50px; padding:50px; background-color: white; }</pre>

Grid Opdracht 2

< Terug

Block 1

Block 2

Block 3

Block 4

Block 5

Block 6

Block 7

Block 8

Block 9

Opdracht 3

- Vier blokken in de hoeken
- Eén blok exact in het midden (rij 3, kolom 3)
- Vier blokken in het midden van elke zijde

Deze opdracht vereist inzicht in expliciete positionering met grid-row en grid-column.

HTML	CSS
<pre><!DOCTYPE html> <html lang="en"> <head> <title>Grid Opdracht 3</title> <meta charset="UTF-8"> <meta name="viewport" content="width=device-width, initial-scale=1"> <link rel="stylesheet" href="../CSS/default.css"> <link rel="stylesheet" href="../CSS/Opdracht%203%20CSS%20Grid.css"> </head> <body> <main> <header>Grid Opdracht 3</header> <nav> <div>&lt; Terug</div> </nav> <div class="container"> <div class="block block1">Block 1</div> <div class="block block2">Block 2</div> <div class="block block3">Block 3</div> <div class="block block4">Block 4</div> <div class="block block5">Block 5</div> <div class="block block6">Block 6</div> <div class="block block7">Block 7</div> <div class="block block8">Block 8</div> <div class="block block9">Block 9</div> </div> </main> </body> </html></pre>	<pre>.container{ background-color: cornflowerblue; display: grid; grid-template-columns: auto auto auto auto auto; grid-template-rows: auto auto auto auto auto; } .container .block{ margin:20px; padding:20px; background-color: white; } .block.block1{ grid-column-start: 1; grid-column-end: 2; grid-row-start: 1; grid-row-end: 2; } .block.block2{ grid-column-start: 2; grid-column-end: 3; grid-row-start: 2; grid-row-end: 3; }</pre>

	<pre>.block.block3{ grid-column-start: 3; grid-column-end: 4; grid-row-start: 3; grid-row-end: 4; } .block.block4{ grid-column-start: 4; grid-column-end: 5; grid-row-start: 4; grid-row-end: 5; } .block.block5{ grid-column-start: 5; grid-column-end: 6; grid-row-start: 5; grid-row-end: 6; } .block.block6{ grid-column-start: 5; grid-column-end: 6; grid-row-start: 1; grid-row-end: 2; } .block.block7{ grid-column-start: 4; grid-column-end: 5; grid-row-start: 2; grid-row-end: 3; } .block.block8{ grid-column-start: 1; grid-column-end: 2; grid-row-start: 5;</pre>
--	---

	<pre>grid-row-end: 6; } .block.block9{ grid-column-start: 2; grid-column-end: 3; grid-row-start: 4; grid-row-end: 5; }</pre>
--	---

Grid Opdracht 3

< Terug

Block 1

Block 2

Block 3

Block 4

Block 5

Block 6

Block 7

Block 8

Block 9

Opdracht 4

Doel

Op grote schermen blijft de kruisvorm behouden. Wanneer het scherm smaller wordt dan 700px, worden de blokken weergegeven in een 3 × 3 grid.

Belangrijk concept

De positie en structuur van grid items kan volledig aangepast worden per breakpoint zonder de HTML aan te passen.

HTML	CSS
<pre><!DOCTYPE html> <html lang="en"> <head> <title>Grid Opdracht 4</title> <meta charset="UTF-8"> <meta name="viewport" content="width=device-width, initial- scale=1"> <link rel="stylesheet" href="../CSS/default.css"> <link rel="stylesheet" href="../CSS/Opdracht%204%20CSS%20Grid.css"> </head> <body> <main> <header>Grid Opdracht 4</header> <nav> <div>&lt; Terug</div> </nav> <div class="container"> <div class="block block1">Block 1</div> <div class="block block2">Block 2</div> <div class="block block3">Block 3</div> <div class="block block4">Block 4</div> <div class="block block5">Block 5</div> <div class="block block6">Block 6</div> <div class="block block7">Block 7</div> <div class="block block8">Block 8</div> <div class="block block9">Block 9</div> </div> </main></pre>	<pre>.container { background-color: cornflowerblue; display: grid; grid-template-columns: auto auto auto; grid-template-rows: auto auto auto; } .container .block{ margin:20px; padding:20px; background-color: white; } .block.block1{ grid-column-start: 1; grid-column-end: 2; grid-row-start: 1; grid-row-end: 2; } .block.block2{ grid-column-start: 1; grid-column-end: 2; grid-row-start: 2; grid-row-end: 3; } .block.block3{</pre>

<pre></body> </html></pre>	<pre>grid-column-start: 1; grid-column-end: 2; grid-row-start: 3; grid-row-end: 4; } .block.block4{ grid-column-start:2 ; grid-column-end: 3; grid-row-start: 1; grid-row-end: 2; } .block.block5{ grid-column-start: 2; grid-column-end: 3; grid-row-start: 2; grid-row-end: 3; } .block.block6{ grid-column-start: 2; grid-column-end: 3; grid-row-start: 3; grid-row-end: 4; } .block.block7{ grid-column-start: 3; grid-column-end: 4; grid-row-start: 1; grid-row-end: 2; } .block.block8{ grid-column-start: 3; grid-column-end: 4; grid-row-start: 2; grid-row-end: 3;</pre>
--	---

```
}
.block.block9{
  grid-column-start: 3;
  grid-column-end: 4;

  grid-row-start: 3;
  grid-row-end: 4;
}

/* Wanneer het scherm kleiner wordt dan
700px moeten de blokken in een 3 x 3 grid
getoond wordenn
*/
@media screen and (min-width: 700px){
  .container{

    grid-template-columns: auto auto auto
    auto auto;
    grid-template-rows: auto auto auto auto
    auto;
  }

  .block.block1{
    grid-column-start: 1;
    grid-column-end: 2;

    grid-row-start: 1;
    grid-row-end: 2;

  }
  .block.block2{
    grid-column-start: 2;
    grid-column-end: 3;

    grid-row-start: 2;
    grid-row-end: 3;

  }
  .block.block3{
```

grid-column-start: 3;
grid-column-end: 4;

grid-row-start: 3;
grid-row-end: 4;

}

.block.block4{

grid-column-start: 4;
grid-column-end: 5;

grid-row-start: 4;
grid-row-end: 5;

}

.block.block5{

grid-column-start: 5;
grid-column-end: 6;

grid-row-start: 5;
grid-row-end: 6;

}

.block.block6{

grid-column-start: 5;
grid-column-end: 6;

grid-row-start: 1;
grid-row-end: 2;

}

.block.block7{

grid-column-start: 4;
grid-column-end: 5;

grid-row-start: 2;
grid-row-end: 3;

}

.block.block8{

grid-column-start: 1;
grid-column-end: 2;

grid-row-start: 5;
grid-row-end: 6;

```
}  
.block.block9{  
  grid-column-start: 2;  
  grid-column-end: 3;  
  
  grid-row-start: 4;  
  grid-row-end: 5;  
  
}  
}
```

Grid Opdracht 4

< Terug

Block 1

Block 6

Block 2

Block 7

Block 3

Block 9

Block 4

Block 8

Block 5

Opdracht 5

Doel

Een complex grid ontwerpen met verschillende blokken die meerdere rijen of kolommen overspannen.

Belangrijke aandachtspunten

- Gebruik van grid-column: span x
- Gebruik van gap in plaats van margins
- Achtergrondkleuren om de witruimte duidelijk zichtbaar te maken

Exameninzicht

Deze oefening test het vermogen om een visuele lay-out te analyseren en correct te vertalen naar een CSS Grid-structuur.

HTML	CSS
<pre><!DOCTYPE html> <html lang="en"> <head> <title>Grid Opdracht 5</title> <meta charset="UTF-8"> <meta name="viewport" content="width=device-width, initial- scale=1"> <link rel="stylesheet" href="../CSS/default.css"> <link rel="stylesheet" href="../CSS/Opdracht%205%20CSS%20Grid.css"> </head> <body> <main> <header>Grid Opdracht 5</header> <nav> <div>&lt; Terug</div> </nav> <div class="container"> <div class="block block1">1</div> <div class="block block2">2</div> <div class="block block3">3</div> <div class="block block4">4</div> <div class="block block5">5</div></pre>	<pre>.container{ background-color: white; display: grid; grid-template-columns: auto auto auto ; grid-template-rows: auto auto auto auto auto ; grid-row-gap: 20px; grid-column-gap: 20px; } .container .block{ padding:20px; color: white; font-size: 4em; font-weight:bold; text-align:center; } .block.block1{ background-color: darkmagenta; grid-column-start: 1;</pre>

<pre><div class="block block6">6</div> <div class="block block7">7</div> <div class="block block8">8</div> </div> </main> </body> </html></pre>	<pre>grid-column-end: 4; grid-row-start: 1; grid-row-end: 2; } .block.block2{ background-color: #726EA6; grid-column-start: 1; grid-column-end: 2; grid-row-start: 2; grid-row-end: 4; } .block.block3{ background-color: cornflowerblue; grid-column-start: 2; grid-column-end: 3; grid-row-start: 2; grid-row-end: 3; } .block.block4{ background-color: lightcoral; grid-column-start: 3; grid-column-end: 4; grid-row-start: 2; grid-row-end: 3; } .block.block5{ background-color: mediumslateblue; grid-column-start: 2; grid-column-end: 4; grid-row-start: 3; grid-row-end: 4; } .block.block6{ background-color: darkmagenta; grid-column-start: 1; grid-column-end: 3;</pre>
---	--

```
grid-row-start: 4;
grid-row-end: 5;

}
.block.block7{
  background-color: #726EA6;
  grid-column-start: 3;
  grid-column-end: 4;

  grid-row-start: 4;
  grid-row-end: 5;

}
.block.block8{
  background-color: cornflowerblue;
  grid-column-start: 1;
  grid-column-end: 4;

  grid-row-start: 5;
  grid-row-end: 6;

}
```

Grid Opdracht 5

< Terug

1

2

3

4

5

6

7

8

Opdracht 6

Doel

Een lay-out maken die zich aanpast aan drie schermgroottes:

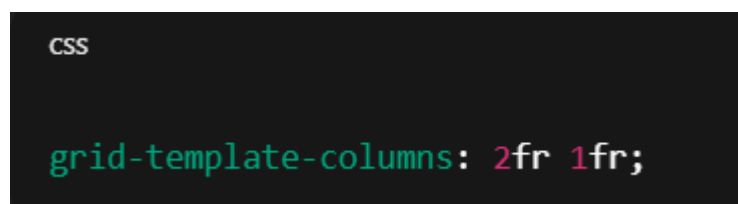
- Breder dan 992px
- Breder dan 772px
- Breder dan 320px

Belangrijke technieken

- Media queries
- Fractional units (fr)
- Wijzigen van de volgorde van blokken via grid-positionering

Fractional units

Voorbeeld:



HTML	CSS
<pre><!DOCTYPE html> <html lang="en"> <head> <title>Grid Opdracht 6</title> <meta charset="UTF-8"> <meta name="viewport" content="width=device-width, initial-scale=1"> <link rel="stylesheet" href="../CSS/default.css"> <link rel="stylesheet" href="../CSS/Opdracht%206%20CSS%20Grid.css"> </head> <body> <main> <div class="block block1">1</div> <div class="block block2">2</div> <div class="block block3">3</div> <div class="block block4">4</div> </main></pre>	<pre>main{ background-color: white; display: grid; grid-template-columns: auto ; grid-template-rows: auto auto auto auto ; grid-row-gap: 20px; grid-column-gap: 20px; } .block{ padding:20px; color: white; font-size: 4em; font-weight:bold; text-align:center;</pre>

<pre></body> </html></pre>	<pre>} .block.block1{ background-color: darkmagenta; grid-column-start: 1; grid-column-end: 2; grid-row-start: 2; grid-row-end: 3; } .block.block2{ background-color: #726EA6; grid-column-start: 1; grid-column-end: 2; grid-row-start: 1; grid-row-end: 2; } .block.block3{ background-color: cornflowerblue; grid-column-start: 1; grid-column-end: 2; grid-row-start: 3; grid-row-end: 4; } .block.block4{ background-color: lightcoral; grid-column-start: 1; grid-column-end: 2; grid-row-start: 4; grid-row-end: 5; } @media screen and (min-width: 772px){ .block.block1{ grid-row-start: 1; grid-row-end: 2; } .block.block2{</pre>
--	---

```
        grid-row-start: 3;
        grid-row-end: 4;

    }
    .block.block3{

        grid-row-start: 2;
        grid-row-end: 3;

    }

}

@media screen and (min-width: 992px){
    .container{

        grid-template-columns: auto auto auto
        ;
        grid-template-rows: auto auto auto ;

    }
    .block.block1{
        grid-column-start: 1;
        grid-column-end: 4;

        grid-row-start: 1;
        grid-row-end: 2;

    }
    .block.block2{
        grid-column-start: 1;
        grid-column-end: 4;

        grid-row-start: 2;
        grid-row-end: 3;

    }
    .block.block3{
        grid-column-start: 1;
        grid-column-end: 2;

        grid-row-start: 3;
        grid-row-end: 4;

    }
}
```

	<pre>.block.block4{ grid-column-start: 2; grid-column-end: 4; grid-row-start: 3; grid-row-end: 4; } </pre>
1 2 3 4	

Opdracht 7

HTML	CSS
<pre><!DOCTYPE html> <html lang="en"> <head> <meta charset="UTF-8"> <title>Demo Grid</title> <link rel="stylesheet" href="../CSS/style.css"> </head> <body> <div> <div class="item-1">1</div> <div class="item-2">2</div> <div class="item-3">3</div> <div class="item-4">4</div> <div class="item-5">5</div> <div class="item-6">6</div> <div class="item-7">7</div> <div class="item-8">8</div> <div class="item-9">9</div> </div> </body> </html></pre>	<pre>/* items */ body div { color: white; font-size: 3em; text-align: center; display: grid; grid-template-rows: auto auto auto auto; grid-template-columns: auto auto; row-gap: 20px; column-gap: 20px; } /* specific item styles */ .item-2 { background: #33a8a5; grid-row-start: 1; grid-row-end: 2; grid-column-start: 2; grid-column-end: 3; } .item-3 { background: #30997a; grid-row-start: 4; grid-row-end: 5; grid-column-start: 1; grid-column-end: 3; } .item-4 { background: #6a478f; grid-row-start: 2; grid-row-end: 3; grid-column-start: 1; grid-column-end: 2; } .item-5 { background: #da6f2b; grid-row-start: 2;</pre>

```
grid-row-end: 3;
grid-column-start: 2;
grid-column-end: 3;
}
```

```
.item-7 {
  background: #e03f3f;
  grid-row-start: 3;
  grid-row-end: 4;
  grid-column-start: 1;
  grid-column-end: 2;
}
```

```
.item-8 {
  background: #59a627;
  grid-row-start: 3;
  grid-row-end: 4;
  grid-column-start: 2;
  grid-column-end: 3;
}
```

```
.item-9 {
  background: #4464a1;
  grid-row-start: 1;
  grid-row-end: 2;
  grid-column-start: 1;
  grid-column-end: 2;
}
```

```
@media screen and (min-width: 760px) {
  body div {
    display: block;
  }
```

```
.item-1 {
  background: #b03532;
}
```

```
.item-6 {
  background: #3d8bb1;
}
```

```
}
```


1

2

3

4

5

6

7